

1. Setup e Imports

```
import json as _json
import json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path
import subprocess

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'matryoshkabatchtopk',
    'tecnica': 'matryoshkabatchtopk',
    'capa': 6,
    'juegos': 12500,
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../../experiments'),
    'extras': {
        'expansion': 16,
        'k': 81,
        'matryoshka_widths': [2048, 4096, 8192],
        'epochs': 1000,
        'patience': 20,
        'lr': 5e-4
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

Metadata de naming:

```
variante: matryoshkabatchtopk
tecnica: matryoshkabatchtopk
capa: 6
juegos: 12500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 16, 'k': 81, 'matryoshka_widths': [2048, 4096, 8192], 'epochs': 1000
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
```

```

    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex3 → {'expansion': 16, 'k': 81, 'matryoshka_widths': [2048, 4096, 8192]}

extras_id: ex3

plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_matryoshka

3. Configuración del SAE

Arquitectura MatryoshkaBatchTopK: - **Nested reconstruction losses:** se calculan pérdidas MSE en múltiples sub-anchos (`matryoshka_widths`) — cada ancho debe reconstruir el input por sí solo - **BatchTopK activation:** threshold global del batch para que el promedio de features activas sea `k` - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con `k` - **Loss** = $\sum \text{MSE}(\text{recon_w}, \text{x})$ para cada ancho `w` `matryoshka_widths`

Parámetro	L1 (standard)	BatchTopK	Matryoshka-BatchTopK
Sparsity	<code>sparsity_coef</code> (soft)	<code>k</code> (promedio del batch)	<code>k</code> (promedio, por nivel)
L0	Variable	Variable por muestra	Variable por muestra, por nivel
Loss	$\text{MSE} + \lambda \cdot \text{L1}$	MSE	$\sum \text{MSE_width}$

Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura `MatryoshkaBatchTopK`.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture': 'matryoshkabatchtopk',
    'd_in': 512,
    'd_sae': 8192,
    'k': 81,
    'matryoshka_widths': [2048, 4096, 8192],
    'lr': 5e-4,
    'train_batch_size_tokens': 4096,
    'hook_name': 'blocks.5.hook_resid_post',
    # P3: betas según config.py de los autores (beta2=0.99 vs default PyTorch 0.999)
    'beta1': 0.9,
    'beta2': 0.99,
    # P1: parámetros para auxiliary loss de dead features (config.py de los autores)
    'n_batches_to_dead': 20, # batches sin activación para declarar feature muerta
    'top_k_aux': 512, # top-k features muertas usadas en L_aux
    'aux_penalty': 1/32, # peso de L_aux (paper ec. 5)
    # P2: normalización del input por muestra
    'input_unit_norm': True,
}

cfg_summary = pd.DataFrame({
    'Parámetro': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'matryoshka_widths',
                  'hook_name', 'lr', 'beta1', 'beta2',
                  'n_batches_to_dead', 'top_k_aux', 'aux_penalty', 'input_unit_norm'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              str(cfg['matryoshka_widths']), cfg['hook_name'], cfg['lr'],
              cfg['beta1'], cfg['beta2'],
              cfg['n_batches_to_dead'], cfg['top_k_aux'], cfg['aux_penalty'],
              cfg['input_unit_norm']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro	Valor
architecture	matryoshkabatchtopk
d_in	512
d_sae	8192
k (L0 promedio)	81
matryoshka_widths	[2048, 4096, 8192]

hook_name	blocks.5.hook_resid_post
lr	0.0005
beta1	0.9
beta2	0.99
n_batches_to_dead	20
top_k_aux	512
aux_penalty	0.03125
input_unit_norm	True

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```
class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path, mmap_mode='r')
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_META['hook']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"], shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=cfg["train_batch_size_tokens"], shuffle=False)

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape': [str(full_dataset.activations.shape), '', ''],
})
```

```
})
print(dataset_info.to_string(index=False))
```

	Conjunto	Muestras	Shape
	Train	590,000	(737500, 512)
	Validación	147,500	
	Total	737,500	

5. Modelo: MatryoshkaBatchTopK SAE

Implementación de la arquitectura MatryoshkaBatchTopK según el paper (ec. 3-5) y el código de los autores (`sae.py`).

Arquitectura: - El encoder proyecta **una sola vez** con W_{enc} completo ($d_{in} \rightarrow d_{sae}$) para obtener $f(x)$ de tamaño d_{sae} (paper ec. 3) - **BatchTopK global único:** se seleccionan los $k \times batch_size$ valores más altos sobre los $d_{sae} \times batch_size$ valores del batch completo — un único threshold compartido entre todos los anchos (paper ec. 6) - Las reconstrucciones por nivel usan **prefijos del mismo vector:** `all_acts_topk[:, 0:w]` — las features 0:2048 son literalmente las mismas en el nivel $w=2048$ y en el nivel $w=4096$ (paper ec. 4) - En **eval**, se usa un threshold persistente calibrado con EMA durante training — no se recalcula (paper ec. 7) - **L0 es variable por muestra pero promedia k en cada batch**

Parámetro	L1 (standard)	BatchTopK	Matryoshka-BatchTopK
Sparsity	<code>sparsity_coef</code> (soft)	k (promedio del batch)	k (promedio, threshold global)
L0	Variable	Variable por muestra	Variable por muestra
Loss	$MSE + \cdot L1$	MSE	$\Sigma MSE_{width} + \cdot L_{aux}$

```
class MatryoshkaBatchTopKSAE(nn.Module):
    """
    Sparse Autoencoder con activación MatryoshkaBatchTopK.
    Corregido según paper (ec. 3-5) e implementación de los autores (sae.py).

    Correcciones aplicadas:
    - P1: encoding único con  $W_{enc}$  completo; BatchTopK global; prefijos compartidos
    - P1: BatchTopK preserva valores originales ( $x * indicator$ ), threshold EMA para eval
```

```

- P1: auxiliary loss para dead features (sae.py:198-221)
- P2: make_decoder_weights_and_grad_unit_norm proyecta gradiente (sae.py:51-57)
- P2: preprocessing input_unit_norm configurable (sae.py:36-43)
- P3: W_dec inicializado como transpuesta de W_enc (sae.py:93-94)
"""
def __init__(self, d_in, d_sae, k, matryoshka_widths,
              n_batches_to_dead=20, top_k_aux=512, aux_penalty=1/32,
              input_unit_norm=False):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.matryoshka_widths = sorted(matryoshka_widths)
    assert self.matryoshka_widths[-1] == d_sae, \
        f"El último ancho ({self.matryoshka_widths[-1]}) debe ser d_sae ({d_sae})"

    self.n_batches_to_dead = n_batches_to_dead
    self.top_k_aux = top_k_aux
    self.aux_penalty = aux_penalty
    self.input_unit_norm = input_unit_norm

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    # P3: W_dec = W_enc^T normalizado (sae.py:93-94) - inicialización tied
    nn.init.kaiming_uniform_(self.W_enc)
    self.W_dec.data[:] = self.W_enc.t().data
    self.W_dec.data[:] = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)

    # P1: threshold persistente calibrado con EMA durante training (sae.py:97)
    self.register_buffer('threshold', torch.tensor(0.0))
    # P1: contador de batches sin activación por feature - para L_aux (sae.py:30-32)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))

# -----
# P2: Preprocessing del input (sae.py:36-48)
# -----

def preprocess_input(self, x):
    if self.input_unit_norm:
        x_mean = x.mean(dim=-1, keepdim=True)

```

```

        x = x - x_mean
        x_std = x.std(dim=-1, keepdim=True)
        x = x / (x_std + 1e-5)
        return x, x_mean, x_std
    return x, None, None

def postprocess_output(self, x_reconstruct, x_mean, x_std):
    if self.input_unit_norm and x_mean is not None:
        x_reconstruct = x_reconstruct * x_std + x_mean
    return x_reconstruct

# -----
# P1: Encoding único + BatchTopK global (sae.py:100-119)
# -----

def compute_activations(self, x_cent):
    # P1: encoding ÚNICO con W_enc completo - un solo pase para todos los anchos
    # Las features 0:w son literalmente las mismas en todos los niveles (paper ec. 3)
    pre_acts = x_cent @ self.W_enc      # [batch, d_sae]
    acts      = F.relu(pre_acts)        # ReLU antes del topk (sae.py:102)

    if self.training:
        # P1: BatchTopK global sobre d_sae*batch_size valores (paper ec. 6)
        # Preserva valores originales con scatter - NO resta threshold (sae.py:104-114)
        topk_result = torch.topk(acts.flatten(), self.k * x_cent.shape[0])
        acts_topk = (
            torch.zeros_like(acts.flatten())
            .scatter(-1, topk_result.indices, topk_result.values)
            .reshape(acts.shape)
        )
        self._update_threshold(acts_topk)
    else:
        # P1: eval usa threshold persistente, no recalcula (paper ec. 7, sae.py:117)
        acts_topk = torch.where(acts > self.threshold, acts, torch.zeros_like(acts))

    return acts, acts_topk

@torch.no_grad()
def _update_threshold(self, acts_topk, lr=0.01):
    # P1: EMA del mínimo valor positivo - calibra para eval (sae.py:224-228)
    positive_mask = acts_topk > 0
    if positive_mask.any():

```



```

        min_positive = acts_topk[positive_mask].min()
        self.threshold = (1 - lr) * self.threshold + lr * min_positive

@torch.no_grad()
def _update_inactive_features(self, acts_topk):
    # P1: necesario para L_aux - trackea qué features no se activan (sae.py:60-62)
    self.num_batches_not_active += (acts_topk.sum(0) == 0).float()
    self.num_batches_not_active[acts_topk.sum(0) > 0] = 0

# -----
# Forward
# -----

def forward(self, x):
    """
    Retorna:
        outputs      : lista de (recon_w, hidden_w) para cada ancho matryoshka
                       recon_w está en espacio normalizado (= espacio original si input_unit_norm=True)
        all_acts      : pre-topk [batch, d_sae] - para L_aux
        all_acts_topk: post-topk [batch, d_sae]
        x_proc        : x preprocesado (= x si input_unit_norm=False)
    """
    x_proc, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_proc - self.b_dec
    all_acts, all_acts_topk = self.compute_activations(x_cent)

    # P1: reconstrucciones por PREFIJO compartido (paper ec. 4, sae.py:149-155)
    # all_acts_topk[:, 0:w] son las mismas features en todos los niveles anidados
    # recon_w se mantiene en espacio normalizado para que la loss compare espacios iguales
    outputs = []
    for w in self.matryoshka_widths:
        hidden_w = all_acts_topk[:, :w]
        recon_w = hidden_w @ self.W_dec[:, :w] + self.b_dec
        outputs.append((recon_w, hidden_w))

    self._update_inactive_features(all_acts_topk)
    return outputs, all_acts, all_acts_topk, x_proc

# -----
# P1: Auxiliary loss para dead features (sae.py:198-221)
# -----

```

```

def get_auxiliary_loss(self, x_proc, x_reconstruct_full, all_acts):
    residual = x_proc.float() - x_reconstruct_full.float()
    aux_reconstruct = torch.zeros_like(residual)

    dead_features = self.num_batches_not_active >= self.n_batches_to_dead

    if dead_features.sum() > 0:
        acts_topk_aux = torch.topk(
            all_acts[:, dead_features],
            min(self.top_k_aux, int(dead_features.sum()))),
            dim=-1,
        )
        acts_aux = torch.zeros_like(all_acts[:, dead_features]).scatter(
            -1, acts_topk_aux.indices, acts_topk_aux.values
        )
        aux_reconstruct = aux_reconstruct + acts_aux @ self.W_dec[dead_features]

    if aux_reconstruct.abs().sum() > 0:
        return self.aux_penalty * (aux_reconstruct.float() - residual.float()).pow(2).mean()
    return torch.tensor(0.0, device=x_proc.device)

# -----
# P2: Constraint de norma unitaria con proyección del gradiente (sae.py:51-57)
# -----

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    # P2: proyecta gradiente al espacio tangente de la esfera unitaria + normaliza
    # @torch.no_grad() evita trazar grafo interno - no impide leer .grad (sae.py:50)
    # Llamar ANTES de optimizer.step() (training.py:31)
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (
        (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    )
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo
model = MatryoshkaBatchTopKSAE(
    d_in=cfg['d_in'],
    d_sae=cfg['d_sae'],

```

```

k=cfg['k'],
matryoshka_widths=cfg['matryoshka_widths'],
n_batches_to_dead=cfg['n_batches_to_dead'],
top_k_aux=cfg['top_k_aux'],
aux_penalty=cfg['aux_penalty'],
input_unit_norm=cfg['input_unit_norm'],
).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Anchos matryoshka',
        'Hidden activos (por nivel)', 'Decoder (W_dec)', 'Output', 'Total params'
    ],
    'Dimensión': [
        f'{cfg["d_in"]}',
        f'{cfg["d_in"]} -> {cfg["d_sae"]} (único pase, prefijos compartidos)',
        f'BatchTopK global (k={cfg["k"]}) - threshold EMA en eval',
        str(cfg['matryoshka_widths']),
        f' {cfg["k"]} activos (promedio del batch)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]} (slices de prefijo)',
        f'{cfg["d_in"]}',
        f'{num_params:,}'
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 8192 (único pase, prefijos compartidos)
Activación	BatchTopK global (k=81) - threshold EMA en eval
Anchos matryoshka	[2048, 4096, 8192]
Hidden activos (por nivel)	81 activos (promedio del batch)
Decoder (W_dec)	8192 -> 512 (slices de prefijo)
Output	512
Total params	8,389,120

6. Función de Pérdida MatryoshkaBatchTopK

Loss = mean MSE(recon_w, x) para cada ancho w matryoshka_widths +
 · L_aux

- **MSE en cada nivel:** error de reconstrucción desde cada sub-espacio matryoshka
- **Sin L1 penalty:** la sparsity está controlada por el threshold global de BatchTopK
- **Loss total = media de MSE en todos los anchos** — mantiene la escala estable y preserva el balance entre reconstrucción y aux_loss independientemente del número de anchos (alineado con Bussmann sae.py:172)
- La loss anidada fuerza a que los sub-espacios más pequeños también sean representaciones completas del input

```
def loss_function(x_proc, outputs, all_acts, model):
    """
    Pérdida MatryoshkaBatchTopK: media de MSE en cada ancho + auxiliary loss.

    Paper ec. 5:  $L(x) = \sum_{m \in M} \|x - \hat{x}\|^2 + \lambda L_{aux}$ 
    Bussmann (sae.py): promedia dividiendo por n_grupos - mantiene la escala de
    la MSE estable independientemente del número de anchos, y preserva el balance
    entre reconstrucción y aux_loss.

    Args:
        x_proc: activaciones preprocesadas [batch, d_in] (= x si input_unit_norm=False)
        outputs: lista de (recon_w, hidden_w) por ancho matryoshka
        all_acts: activaciones pre-topk [batch, d_sae] - para L_aux
        model: referencia al modelo (para get_auxiliary_loss)

    Returns:
        total_loss, mse_per_width, aux_loss
    """
    x_reconstruct_full = outputs[-1][0] # reconstrucción del ancho máximo

    # Media de MSE en todos los anchos + término b_dec (alineado con Bussmann sae.py:163-172)
    mse_bdec = F.mse_loss(model.b_dec.expand_as(x_proc), x_proc)
    mse_per_width = [F.mse_loss(recon_w, x_proc) for recon_w, _ in outputs]
    mean_mse = (mse_bdec + sum(mse_per_width)) / (len(mse_per_width) + 1)

    # P1: auxiliary loss para dead features (paper ec. 5:  $\lambda L_{aux}$ , sae.py:198-221)
    aux_loss = model.get_auxiliary_loss(x_proc, x_reconstruct_full, all_acts)

    return mean_mse + aux_loss, mse_per_width, aux_loss

# P3: betas según config.py de los autores - beta2=0.99 vs default PyTorch 0.999
# (config.py: beta1=0.9, beta2=0.99)
optimizer = torch.optim.Adam(
    model.parameters(),
    lr=cfg["lr"],
```

```

    betas=(cfg["beta1"], cfg["beta2"])
)

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'lr', 'beta1', 'beta2',
                  'k (L0 promedio)', 'Anchos matryoshka',
                  'aux_penalty', 'n_batches_to_dead', 'top_k_aux'],
    'Valor':      ['Adam', cfg["lr"], cfg["beta1"], cfg["beta2"],
                  cfg["k"], str(cfg['matryoshka_widths']),
                  cfg['aux_penalty'], cfg['n_batches_to_dead'], cfg['top_k_aux']]
})
print(optim_info.to_string(index=False))

```

Parámetro	Valor
Optimizador	Adam
lr	0.0005
beta1	0.9
beta2	0.99
k (L0 promedio)	81
Anchos matryoshka	[2048, 4096, 8192]
aux_penalty	0.03125
n_batches_to_dead	20
top_k_aux	512

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']
WIDTHS    = cfg['matryoshka_widths']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)
BEST_CKPT = save_dir / best_model_name

```

```

history = {
    'train_total_loss': [],
    'train_aux_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_aux_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}
for w in WIDTHS:
    history[f'val_mse_w{w}'] = []

best_val_loss = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'Anchos matryoshka: {WIDTHS}')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # ----- TRAIN
    model.train()
    t_total = t_aux = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()

        # P1: forward devuelve (outputs, all_acts, all_acts_topk, x_proc)
        outputs, all_acts, all_acts_topk, x_proc = model(x)
        total_loss, _, aux_loss_val = loss_function(x_proc, outputs, all_acts, model)
        total_loss.backward()

        # P2: proyectar gradiente al espacio tangente de la esfera + normalizar
        # ANTES de optimizer.step() (sae.py:51-57, training.py:31)
        model.make_decoder_weights_and_grad_unit_norm()

```

```

optimizer.step()
# REMOVED: model.normalize_decoder() - reemplazado por make_decoder_weights_and_grad

with torch.no_grad():
    hidden_full = outputs[-1][1]
    t_total += total_loss.item()
    t_aux    += aux_loss_val.item()
    t_l0     += (hidden_full > 0).float().sum(dim=1).mean().item()
    t_frac   += (hidden_full > 0).float().mean().item()

n = len(train_loader)
history['train_total_loss'].append(t_total / n)
history['train_aux_loss'].append(t_aux / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# ----- VALIDATION
model.eval()
v_total = v_aux = v_l0 = v_frac = 0.0
v_per_width = {w: 0.0 for w in WIDTHHS}

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        outputs, all_acts, all_acts_topk, x_proc = model(x)
        total_loss, mse_per_width, aux_loss_val = loss_function(x_proc, outputs, all_acts)
        hidden_full = outputs[-1][1]

        v_total += total_loss.item()
        v_aux    += aux_loss_val.item()
        v_l0     += (hidden_full > 0).float().sum(dim=1).mean().item()
        v_frac   += (hidden_full > 0).float().mean().item()
        for w, mse_w in zip(WIDTHHS, mse_per_width):
            v_per_width[w] += mse_w.item()

nv = len(val_loader)
history['val_total_loss'].append(v_total / nv)
history['val_aux_loss'].append(v_aux / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)
for w in WIDTHHS:
    history[f'val_mse_w{w}'].append(v_per_width[w] / nv)

```

```

width_str = ' | '.join(f'w{w}={history[f"val_mse_w{w}"][-1]:.4f}' for w in WIDTHS)
print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train: {history['train_total_loss'][-1]:.6f} | "
      f"Val: {history['val_total_loss'][-1]:.6f} | "
      f"L_aux: {history['val_aux_loss'][-1]:.6f} | "
      f"L0: {history['val_l0_sparsity'][-1]:.1f} | "
      f"{width_str}")

# Early stopping (basado en loss total de validación)
if history['val_total_loss'][-1] < best_val_loss - 1e-4:
    best_val_loss = history['val_total_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0
    torch.save(model.state_dict(), BEST_CKPT)
    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f' Mejor modelo guardado (época {best_epoch} | Val Loss: {best_val_loss:.6f})

else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val Loss: {best_val_loss:.6f}')

```

Iniciando entrenamiento: 1000 épocas máximo
 Early stopping: patience=20

k = 81 features activas (L0 promedio k por batch)

Anchos matryoshka: [2048, 4096, 8192]

Best checkpoint: saved_model\sae_matryoshkabatchtopk_matryoshkabatchtopk_l6_12500g_ex3_best.

=====

Época	1/1000	Train: 0.550663	Val: 0.553672	L_aux: 0.000000	L0: 218.3	w2048=0.3298
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 1 Val Loss: 0.553672)						
Época	2/1000	Train: 0.474566	Val: 0.456195	L_aux: 0.000000	L0: 90.0	w2048=0.3098
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 2 Val Loss: 0.456195)						
Época	3/1000	Train: 0.453401	Val: 0.451975	L_aux: 0.000000	L0: 79.0	w2048=0.3061
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 3 Val Loss: 0.451975)						
Época	4/1000	Train: 0.441476	Val: 0.443998	L_aux: 0.000000	L0: 78.3	w2048=0.2960
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 4 Val Loss: 0.443998)						
Época	5/1000	Train: 0.433652	Val: 0.437462	L_aux: 0.000000	L0: 78.6	w2048=0.2876
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 5 Val Loss: 0.437462)						
Época	6/1000	Train: 0.427955	Val: 0.432794	L_aux: 0.000000	L0: 78.7	w2048=0.2815
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 6 Val Loss: 0.432794)						
Época	7/1000	Train: 0.423639	Val: 0.428594	L_aux: 0.000000	L0: 79.2	w2048=0.2757
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 7 Val Loss: 0.428594)						
Época	8/1000	Train: 0.420105	Val: 0.425817	L_aux: 0.000000	L0: 79.1	w2048=0.2720
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 8 Val Loss: 0.425817)						
Época	9/1000	Train: 0.417250	Val: 0.423209	L_aux: 0.000000	L0: 79.2	w2048=0.2684
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 9 Val Loss: 0.423209)						
Época	10/1000	Train: 0.414762	Val: 0.420852	L_aux: 0.000000	L0: 79.4	w2048=0.2651
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 10 Val Loss: 0.420852)						
Época	11/1000	Train: 0.412662	Val: 0.418879	L_aux: 0.000000	L0: 79.5	w2048=0.2625
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 11 Val Loss: 0.418879)						
Época	12/1000	Train: 0.410770	Val: 0.417200	L_aux: 0.000000	L0: 79.5	w2048=0.2602
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 12 Val Loss: 0.417200)						
Época	13/1000	Train: 0.409075	Val: 0.415919	L_aux: 0.000000	L0: 79.2	w2048=0.2585
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma						
Mejor modelo guardado (época 13 Val Loss: 0.415919)						

Época 14/1000 | Train: 0.407683 | Val: 0.414360 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2564
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 14 | Val Loss: 0.414360)
 Época 15/1000 | Train: 0.406394 | Val: 0.413531 | L_aux: 0.000000 | L0: 79.2 | w2048=0.2554
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 15 | Val Loss: 0.413531)
 Época 16/1000 | Train: 0.405285 | Val: 0.412701 | L_aux: 0.000000 | L0: 79.0 | w2048=0.2543
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 16 | Val Loss: 0.412701)
 Época 17/1000 | Train: 0.404282 | Val: 0.411702 | L_aux: 0.000000 | L0: 79.2 | w2048=0.2532
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 17 | Val Loss: 0.411702)
 Época 18/1000 | Train: 0.403462 | Val: 0.410935 | L_aux: 0.000000 | L0: 79.2 | w2048=0.2524
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 18 | Val Loss: 0.410935)
 Época 19/1000 | Train: 0.402631 | Val: 0.410343 | L_aux: 0.000000 | L0: 79.2 | w2048=0.2517
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 19 | Val Loss: 0.410343)
 Época 20/1000 | Train: 0.401989 | Val: 0.409815 | L_aux: 0.000000 | L0: 79.2 | w2048=0.2511
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 20 | Val Loss: 0.409815)
 Época 21/1000 | Train: 0.401385 | Val: 0.409122 | L_aux: 0.000000 | L0: 79.4 | w2048=0.2504
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 21 | Val Loss: 0.409122)
 Época 22/1000 | Train: 0.400828 | Val: 0.408560 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2498
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 22 | Val Loss: 0.408560)
 Época 23/1000 | Train: 0.400241 | Val: 0.408270 | L_aux: 0.000000 | L0: 79.4 | w2048=0.2495
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 23 | Val Loss: 0.408270)
 Época 24/1000 | Train: 0.399792 | Val: 0.407710 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2489
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 24 | Val Loss: 0.407710)
 Época 25/1000 | Train: 0.399426 | Val: 0.407259 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2484
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 25 | Val Loss: 0.407259)
 Época 26/1000 | Train: 0.398948 | Val: 0.407064 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2483
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 26 | Val Loss: 0.407064)
 Época 27/1000 | Train: 0.398582 | Val: 0.406890 | L_aux: 0.000000 | L0: 79.4 | w2048=0.2482
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 27 | Val Loss: 0.406890)
 Época 28/1000 | Train: 0.398235 | Val: 0.406310 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2474

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 28 | Val Loss: 0.406310)
Época 29/1000 | Train: 0.397903 | Val: 0.406065 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2473
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 29 | Val Loss: 0.406065)
Época 30/1000 | Train: 0.397605 | Val: 0.405771 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2469
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 30 | Val Loss: 0.405771)
Época 31/1000 | Train: 0.397329 | Val: 0.405527 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2466
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 31 | Val Loss: 0.405527)
Época 32/1000 | Train: 0.397004 | Val: 0.405115 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2461
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 32 | Val Loss: 0.405115)
Época 33/1000 | Train: 0.396704 | Val: 0.405347 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2465
Época 34/1000 | Train: 0.396503 | Val: 0.404938 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2459
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 34 | Val Loss: 0.404938)
Época 35/1000 | Train: 0.396272 | Val: 0.404498 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2454
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 35 | Val Loss: 0.404498)
Época 36/1000 | Train: 0.396018 | Val: 0.404387 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2453
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 36 | Val Loss: 0.404387)
Época 37/1000 | Train: 0.395784 | Val: 0.404499 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2456
Época 38/1000 | Train: 0.395605 | Val: 0.404142 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2451
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 38 | Val Loss: 0.404142)
Época 39/1000 | Train: 0.395389 | Val: 0.404061 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2450
Época 40/1000 | Train: 0.395141 | Val: 0.403917 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2449
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 40 | Val Loss: 0.403917)
Época 41/1000 | Train: 0.394957 | Val: 0.403764 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2448
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 41 | Val Loss: 0.403764)
Época 42/1000 | Train: 0.394815 | Val: 0.403476 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2444
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 42 | Val Loss: 0.403476)
Época 43/1000 | Train: 0.394634 | Val: 0.403550 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2445
Época 44/1000 | Train: 0.394435 | Val: 0.403230 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2441
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
Mejor modelo guardado (época 44 | Val Loss: 0.403230)
Época 45/1000 | Train: 0.394272 | Val: 0.402912 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2437

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 45 | Val Loss: 0.402912)

Época 46/1000 | Train: 0.394100 | Val: 0.402818 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2436

Época 47/1000 | Train: 0.393964 | Val: 0.402862 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2436

Época 48/1000 | Train: 0.393794 | Val: 0.402406 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2430

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 48 | Val Loss: 0.402406)

Época 49/1000 | Train: 0.393609 | Val: 0.402534 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2433

Época 50/1000 | Train: 0.393463 | Val: 0.402606 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2434

Época 51/1000 | Train: 0.393360 | Val: 0.402276 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2429

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 51 | Val Loss: 0.402276)

Época 52/1000 | Train: 0.393197 | Val: 0.402303 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2430

Época 53/1000 | Train: 0.393090 | Val: 0.401972 | L_aux: 0.000000 | L0: 80.3 | w2048=0.2426

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 53 | Val Loss: 0.401972)

Época 54/1000 | Train: 0.392975 | Val: 0.401928 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2425

Época 55/1000 | Train: 0.392793 | Val: 0.402341 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2432

Época 56/1000 | Train: 0.392722 | Val: 0.401792 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2424

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 56 | Val Loss: 0.401792)

Época 57/1000 | Train: 0.392592 | Val: 0.402139 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2428

Época 58/1000 | Train: 0.392432 | Val: 0.401899 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2426

Época 59/1000 | Train: 0.392288 | Val: 0.401703 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2422

Época 60/1000 | Train: 0.392268 | Val: 0.401623 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2421

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 60 | Val Loss: 0.401623)

Época 61/1000 | Train: 0.392097 | Val: 0.401764 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2424

Época 62/1000 | Train: 0.392103 | Val: 0.401129 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2414

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 62 | Val Loss: 0.401129)

Época 63/1000 | Train: 0.391943 | Val: 0.400853 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2410

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 63 | Val Loss: 0.400853)

Época 64/1000 | Train: 0.391888 | Val: 0.400890 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2411

Época 65/1000 | Train: 0.391771 | Val: 0.400994 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2412

Época 66/1000 | Train: 0.391624 | Val: 0.401543 | L_aux: 0.000000 | L0: 79.4 | w2048=0.2421

Época 67/1000 | Train: 0.391531 | Val: 0.400857 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2411

Época 68/1000 | Train: 0.391436 | Val: 0.401482 | L_aux: 0.000000 | L0: 79.3 | w2048=0.2420

Época 69/1000 | Train: 0.391363 | Val: 0.400769 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2410

Época 70/1000 | Train: 0.391286 | Val: 0.400782 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2409

Época 71/1000 | Train: 0.391209 | Val: 0.400622 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2407

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 71 | Val Loss: 0.400622)

Época 72/1000 | Train: 0.391181 | Val: 0.400083 | L_aux: 0.000000 | L0: 80.8 | w2048=0.2399

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 72 | Val Loss: 0.400083)

Época 73/1000 | Train: 0.391108 | Val: 0.400402 | L_aux: 0.000000 | L0: 80.3 | w2048=0.2404

Época 74/1000 | Train: 0.391012 | Val: 0.400309 | L_aux: 0.000000 | L0: 80.3 | w2048=0.2402

Época 75/1000 | Train: 0.390908 | Val: 0.400692 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2408

Época 76/1000 | Train: 0.390801 | Val: 0.400678 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2408

Época 77/1000 | Train: 0.390782 | Val: 0.400319 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2403

Época 78/1000 | Train: 0.390703 | Val: 0.400500 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2406

Época 79/1000 | Train: 0.390635 | Val: 0.400224 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2402

Época 80/1000 | Train: 0.390553 | Val: 0.399866 | L_aux: 0.000000 | L0: 80.5 | w2048=0.2396

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 80 | Val Loss: 0.399866)

Época 81/1000 | Train: 0.390491 | Val: 0.400460 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2405

Época 82/1000 | Train: 0.390395 | Val: 0.400530 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2406

Época 83/1000 | Train: 0.390314 | Val: 0.400639 | L_aux: 0.000000 | L0: 79.4 | w2048=0.2407

Época 84/1000 | Train: 0.390296 | Val: 0.399867 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2396

Época 85/1000 | Train: 0.390185 | Val: 0.400763 | L_aux: 0.000000 | L0: 79.1 | w2048=0.2409

Época 86/1000 | Train: 0.390198 | Val: 0.400149 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2400

Época 87/1000 | Train: 0.390145 | Val: 0.400295 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2402

Época 88/1000 | Train: 0.390099 | Val: 0.400009 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2397

Época 89/1000 | Train: 0.390021 | Val: 0.400128 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2399

Época 90/1000 | Train: 0.389992 | Val: 0.399214 | L_aux: 0.000000 | L0: 80.7 | w2048=0.2386

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 90 | Val Loss: 0.399214)

Época 91/1000 | Train: 0.389917 | Val: 0.399769 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2394

Época 92/1000 | Train: 0.389866 | Val: 0.400015 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2398

Época 93/1000 | Train: 0.389846 | Val: 0.399240 | L_aux: 0.000000 | L0: 80.6 | w2048=0.2385

Época 94/1000 | Train: 0.389755 | Val: 0.400402 | L_aux: 0.000000 | L0: 79.1 | w2048=0.2402

Época 95/1000 | Train: 0.389736 | Val: 0.399712 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2393

Época 96/1000 | Train: 0.389708 | Val: 0.399497 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2390

Época 97/1000 | Train: 0.389605 | Val: 0.399386 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2388

Época 98/1000 | Train: 0.389565 | Val: 0.400056 | L_aux: 0.000000 | L0: 79.4 | w2048=0.2397

Época 99/1000 | Train: 0.389486 | Val: 0.399437 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2388

Época 100/1000 | Train: 0.389470 | Val: 0.399333 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2386

Época 101/1000 | Train: 0.389432 | Val: 0.399409 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2388

Época 102/1000 | Train: 0.389384 | Val: 0.399361 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2387

Época 103/1000 | Train: 0.389325 | Val: 0.399617 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2391

Época 104/1000 | Train: 0.389287 | Val: 0.399400 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2387

Época 105/1000 | Train: 0.389191 | Val: 0.399682 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2391

Época 106/1000 | Train: 0.389173 | Val: 0.399633 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2390

Época 107/1000 | Train: 0.389185 | Val: 0.398635 | L_aux: 0.000000 | L0: 80.8 | w2048=0.2376

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 107 | Val Loss: 0.398635)

Época 108/1000		Train: 0.389357		Val: 0.399744		L_aux: 0.000000		L0: 79.3		w2048=0.2392
Época 109/1000		Train: 0.389039		Val: 0.399831		L_aux: 0.000000		L0: 79.2		w2048=0.2393
Época 110/1000		Train: 0.389043		Val: 0.399033		L_aux: 0.000000		L0: 80.1		w2048=0.2381
Época 111/1000		Train: 0.388970		Val: 0.399775		L_aux: 0.000000		L0: 79.2		w2048=0.2392
Época 112/1000		Train: 0.388986		Val: 0.399031		L_aux: 0.000000		L0: 80.0		w2048=0.2381
Época 113/1000		Train: 0.389824		Val: 0.398938		L_aux: 0.000000		L0: 80.1		w2048=0.2380
Época 114/1000		Train: 0.388834		Val: 0.399800		L_aux: 0.000000		L0: 79.1		w2048=0.2392
Época 115/1000		Train: 0.388887		Val: 0.399010		L_aux: 0.000000		L0: 80.0		w2048=0.2380
Época 116/1000		Train: 0.388771		Val: 0.399482		L_aux: 0.000000		L0: 79.3		w2048=0.2387
Época 117/1000		Train: 0.388787		Val: 0.398891		L_aux: 0.000000		L0: 80.1		w2048=0.2379
Época 118/1000		Train: 0.388722		Val: 0.399107		L_aux: 0.000000		L0: 79.8		w2048=0.2381
Época 119/1000		Train: 0.388719		Val: 0.399044		L_aux: 0.000000		L0: 79.8		w2048=0.2380
Época 120/1000		Train: 0.388717		Val: 0.398477		L_aux: 0.000000		L0: 80.5		w2048=0.2372

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 120 | Val Loss: 0.398477)

Época 121/1000		Train: 0.388681		Val: 0.398227		L_aux: 0.000000		L0: 80.7		w2048=0.2368
----------------	--	-----------------	--	---------------	--	-----------------	--	----------	--	--------------

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 121 | Val Loss: 0.398227)

Época 122/1000		Train: 0.388593		Val: 0.399182		L_aux: 0.000000		L0: 79.5		w2048=0.2382
Época 123/1000		Train: 0.388571		Val: 0.399200		L_aux: 0.000000		L0: 79.4		w2048=0.2382
Época 124/1000		Train: 0.388587		Val: 0.398329		L_aux: 0.000000		L0: 80.5		w2048=0.2369
Época 125/1000		Train: 0.388524		Val: 0.398636		L_aux: 0.000000		L0: 80.1		w2048=0.2374
Época 126/1000		Train: 0.388467		Val: 0.398846		L_aux: 0.000000		L0: 79.8		w2048=0.2377
Época 127/1000		Train: 0.388464		Val: 0.398778		L_aux: 0.000000		L0: 79.8		w2048=0.2375
Época 128/1000		Train: 0.388443		Val: 0.398192		L_aux: 0.000000		L0: 80.6		w2048=0.2367
Época 129/1000		Train: 0.388395		Val: 0.398758		L_aux: 0.000000		L0: 79.8		w2048=0.2375
Época 130/1000		Train: 0.388329		Val: 0.399096		L_aux: 0.000000		L0: 79.4		w2048=0.2380
Época 131/1000		Train: 0.388298		Val: 0.399054		L_aux: 0.000000		L0: 79.4		w2048=0.2379
Época 132/1000		Train: 0.388233		Val: 0.400069		L_aux: 0.000000		L0: 78.2		w2048=0.2393
Época 133/1000		Train: 0.388282		Val: 0.398472		L_aux: 0.000000		L0: 80.0		w2048=0.2371
Época 134/1000		Train: 0.388201		Val: 0.398306		L_aux: 0.000000		L0: 80.3		w2048=0.2368
Época 135/1000		Train: 0.388770		Val: 0.398359		L_aux: 0.000000		L0: 80.1		w2048=0.2369
Época 136/1000		Train: 0.388922		Val: 0.398058		L_aux: 0.000000		L0: 80.5		w2048=0.2364

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

Mejor modelo guardado (época 136 | Val Loss: 0.398058)

Época 137/1000		Train: 0.388136		Val: 0.398001		L_aux: 0.000000		L0: 80.5		w2048=0.2363
Época 138/1000		Train: 0.388163		Val: 0.398063		L_aux: 0.000000		L0: 80.5		w2048=0.2364
Época 139/1000		Train: 0.388070		Val: 0.398572		L_aux: 0.000000		L0: 79.8		w2048=0.2371
Época 140/1000		Train: 0.388088		Val: 0.398356		L_aux: 0.000000		L0: 80.0		w2048=0.2368
Época 141/1000		Train: 0.388022		Val: 0.398473		L_aux: 0.000000		L0: 79.8		w2048=0.2369
Época 142/1000		Train: 0.388003		Val: 0.398761		L_aux: 0.000000		L0: 79.4		w2048=0.2373

Época 143/1000 | Train: 0.388019 | Val: 0.398090 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2364
 Época 144/1000 | Train: 0.387978 | Val: 0.397999 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2362
 Época 145/1000 | Train: 0.387877 | Val: 0.398637 | L_aux: 0.000000 | L0: 79.5 | w2048=0.2371
 Época 146/1000 | Train: 0.387874 | Val: 0.398262 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2366
 Época 147/1000 | Train: 0.387860 | Val: 0.397846 | L_aux: 0.000000 | L0: 80.5 | w2048=0.2360
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 147 | Val Loss: 0.397846)
 Época 148/1000 | Train: 0.387852 | Val: 0.398475 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2368
 Época 149/1000 | Train: 0.387811 | Val: 0.398427 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2368
 Época 150/1000 | Train: 0.387799 | Val: 0.398175 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2364
 Época 151/1000 | Train: 0.387756 | Val: 0.398126 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2363
 Época 152/1000 | Train: 0.387774 | Val: 0.397953 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2360
 Época 153/1000 | Train: 0.387730 | Val: 0.398013 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2361
 Época 154/1000 | Train: 0.387695 | Val: 0.398436 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2367
 Época 155/1000 | Train: 0.387682 | Val: 0.397550 | L_aux: 0.000000 | L0: 80.7 | w2048=0.2354
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 155 | Val Loss: 0.397550)
 Época 156/1000 | Train: 0.387648 | Val: 0.398338 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2365
 Época 157/1000 | Train: 0.387586 | Val: 0.398247 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2364
 Época 158/1000 | Train: 0.387578 | Val: 0.397675 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2356
 Época 159/1000 | Train: 0.387543 | Val: 0.398289 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2364
 Época 160/1000 | Train: 0.387543 | Val: 0.397441 | L_aux: 0.000000 | L0: 80.7 | w2048=0.2352
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 160 | Val Loss: 0.397441)
 Época 161/1000 | Train: 0.387499 | Val: 0.398133 | L_aux: 0.000000 | L0: 79.8 | w2048=0.2362
 Época 162/1000 | Train: 0.387466 | Val: 0.398186 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2362
 Época 163/1000 | Train: 0.387476 | Val: 0.397642 | L_aux: 0.000000 | L0: 80.3 | w2048=0.2354
 Época 164/1000 | Train: 0.387406 | Val: 0.398060 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2360
 Época 165/1000 | Train: 0.387374 | Val: 0.398187 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2362
 Época 166/1000 | Train: 0.387422 | Val: 0.397208 | L_aux: 0.000000 | L0: 80.8 | w2048=0.2348
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma
 Mejor modelo guardado (época 166 | Val Loss: 0.397208)
 Época 167/1000 | Train: 0.387356 | Val: 0.398064 | L_aux: 0.000000 | L0: 79.7 | w2048=0.2360
 Época 168/1000 | Train: 0.387340 | Val: 0.397511 | L_aux: 0.000000 | L0: 80.4 | w2048=0.2352
 Época 169/1000 | Train: 0.387359 | Val: 0.397221 | L_aux: 0.000000 | L0: 80.8 | w2048=0.2347
 Época 170/1000 | Train: 0.387294 | Val: 0.397831 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2356
 Época 171/1000 | Train: 0.387273 | Val: 0.397823 | L_aux: 0.000000 | L0: 79.9 | w2048=0.2356
 Época 172/1000 | Train: 0.387236 | Val: 0.397668 | L_aux: 0.000000 | L0: 80.1 | w2048=0.2354
 Época 173/1000 | Train: 0.387203 | Val: 0.398055 | L_aux: 0.000000 | L0: 79.6 | w2048=0.2359
 Época 174/1000 | Train: 0.387201 | Val: 0.397687 | L_aux: 0.000000 | L0: 80.0 | w2048=0.2353
 Época 175/1000 | Train: 0.387218 | Val: 0.397219 | L_aux: 0.000000 | L0: 80.6 | w2048=0.2347
 Época 176/1000 | Train: 0.387190 | Val: 0.397408 | L_aux: 0.000000 | L0: 80.3 | w2048=0.2349
 Época 177/1000 | Train: 0.388431 | Val: 0.397412 | L_aux: 0.000000 | L0: 80.2 | w2048=0.2349

Época 178/1000	Train: 0.387800	Val: 0.397945	L_aux: 0.000000	L0: 79.6	w2048=0.2356
Época 179/1000	Train: 0.387013	Val: 0.398012	L_aux: 0.000000	L0: 79.5	w2048=0.2357
Época 180/1000	Train: 0.387134	Val: 0.397491	L_aux: 0.000000	L0: 80.2	w2048=0.2350
Época 181/1000	Train: 0.387071	Val: 0.397367	L_aux: 0.000000	L0: 80.3	w2048=0.2348
Época 182/1000	Train: 0.387024	Val: 0.397462	L_aux: 0.000000	L0: 80.2	w2048=0.2349
Época 183/1000	Train: 0.386964	Val: 0.397933	L_aux: 0.000000	L0: 79.5	w2048=0.2355
Época 184/1000	Train: 0.386965	Val: 0.397520	L_aux: 0.000000	L0: 80.0	w2048=0.2350
Época 185/1000	Train: 0.386941	Val: 0.397760	L_aux: 0.000000	L0: 79.7	w2048=0.2353
Época 186/1000	Train: 0.386952	Val: 0.397918	L_aux: 0.000000	L0: 79.4	w2048=0.2355

Early stopping en época 186 (mejor: época 166)

Mejor época: 166 | Mejor Val Loss: 0.397208

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg['architecture'],
        'd_in': cfg['d_in'],
        'd_sae': cfg['d_sae'],
        'k': cfg['k'],
        'matryoshka_widths': cfg['matryoshka_widths'],
        'lr': cfg['lr'],
        'hook_name': cfg['hook_name'],
    },
    'history': history,
}

final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)
checkpoint_path = save_dir / final_model_name
```



```

torch.save(checkpoint, checkpoint_path)
print(f'Modelo guardado: {checkpoint_path}')

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_matryoshkabatchtopk_matryoshkabatchtopk_l6_12500g_ex3_final
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_ma

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_matryoshkab

9. Métricas básicas del modelo

```

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción (loss total = media de MSE en todos los anchos)
    'val_total_loss_final': history['val_total_loss'][-1],
    'val_total_loss_best': best_val_loss,
    # MSE por ancho matryoshka (último epoch)
    **{f'val_mse_w{w}_final': history[f'val_mse_w{w}'][-1] for w in WIDTHS},
    # Sparsity (L0 del ancho máximo)

```

```

'val_l0_final':          history['val_l0_sparsity'][-1],
'val_fraction_active_final': history['val_fraction_active'][-1],
# Contexto del entrenamiento
'best_epoch':           best_epoch,
'total_epochs':         len(history['train_total_loss']),
'train_total_loss_final': history['train_total_loss'][-1],
'train_l0_final':        history['train_l0_sparsity'][-1],
# Hiperparámetros MatryoshkaBatchTopK
'architecture':         cfg['architecture'],
'k':                    cfg['k'],
'matryoshka_widths':     cfg['matryoshka_widths'],
'd_sae':                cfg['d_sae'],
'd_in':                 cfg['d_in'],
'expansion_factor':      cfg['d_sae'] // cfg['d_in'],
'hook_name':            cfg['hook_name'],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\s

Métrica	Valor
val_total_loss_final	0.397918
val_total_loss_best	0.397208
val_mse_w2048_final	0.235475
val_mse_w4096_final	0.209236
val_mse_w8192_final	0.193423
val_l0_final	79.431684
val_fraction_active_final	0.009696
best_epoch	166
total_epochs	186
train_total_loss_final	0.386952
train_l0_final	81.0
architecture	matryoshkabatchtopk
k	81

matryoshka_widths	[2048, 4096, 8192]
d_sae	8192
d_in	512
expansion_factor	16
hook_name	blocks.5.hook_resid_post

10. Visualización de Resultados

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Total Nested Loss
axes[0].plot(history['train_total_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_total_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title(f'Total Nested Loss ( $\Sigma$  MSE en {len(WIDTHS)} anchos)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('Loss total')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# L0 sparsity (ancho máximo, L0 promedio k)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8,
                label=f'k={cfg["k"]} (promedio objetivo)')
axes[1].set_title(f'L0 Sparsity - ancho máximo (w={cfg["d_sae"]}, promedio k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# MSE por ancho matryoshka (validación)
colors_w = plt.cm.viridis([i / len(WIDTHS) for i in range(len(WIDTHS))])
for w, color in zip(WIDTHS, colors_w):
    axes[2].plot(history[f'val_mse_w{w}'], label=f'w={w}', color=color, alpha=0.85)
if best_epoch > 0:
    axes[2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[2].set_title('Val MSE por Ancho Matryoshka')
```

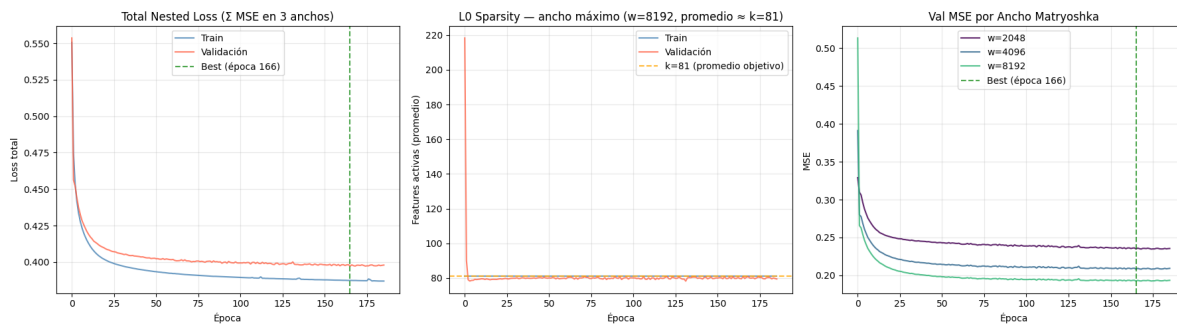
```

axes[2].set_xlabel('Época')
axes[2].set_ylabel('MSE')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)

```



=====

Gráficas guardadas exitosamente

=====

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

```

```

)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_matryoska-batch-topk.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\lay